

Bond-Characteristics Structural Basis Adjustment

A Standalone Technical Specification for Training the Nuisance Model $C_{i,t}$ and Integrating It into a Corporate-Bond Relative-Value Score

Implementation specification

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Abstract

This document specifies one narrow component of a corporate-bond relative-value system: the estimation of a bond-characteristics structural basis $C_{i,t}$. The existing infrastructure is assumed to provide a par-implied survival curve that was fitted using all eligible bonds, including the target bond. We do not refit that curve and we do not attempt to create a synthetic leave-one-out curve. Instead, for each bond we first compute an inclusive curve-pricing basis $\delta_{i,t}^{inc}$, then estimate the bond's own historical normal basis $B_{i,t}^{own}$ from lagged observations. The regression target for the characteristics model is $Y_{i,t}^C = B_{i,t}^{own}$, not future return and not today's raw curve residual. The fitted prediction $C_{i,t} = \hat{\mathbb{E}}[B_{i,t}^{own} \mid q_{i,t}]$ is a nuisance estimate of where a bond with characteristics $q_{i,t}$ normally trades relative to the existing curve. It is then blended with the bond's own history using reliability weights. The result is subtracted from the current inclusive basis before the final RV score is standardized. Every step below is point-in-time and implementable without changing the survival-curve engine.

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1 The one-sentence answer

The symbol $C_{i,t}$ is *not* a training target. It is the fitted prediction from a separate nuisance regression. The target used to train that regression is the bond’s lagged, robust estimate of its normal historical basis to the same inclusive survival curve:

$$Y_{i,t}^C = B_{i,t}^{own}. \quad (1)$$

The characteristics model is then

$$C_{i,t} = \hat{n}_t(q_{i,t}) \approx \mathbb{E}[B_{i,t}^{own} \mid q_{i,t}]. \quad (2)$$

Its purpose is to estimate persistent, non-alpha bond-specific basis. It must not be trained to predict future returns.

2 Scope and non-scope

2.1 What this component does

For a target bond i at time t , this component answers:

Given the bond’s structural characteristics and the way this bond/curve family normally behaves, how many basis points rich or cheap to the existing survival curve would be considered normal rather than a temporary relative-value dislocation?

2.2 What this component does not do

It does not predict five- or ten-day returns. It does not use CDS, equity, RFQ imbalance, order flow, dealer inventory, ETF flows, news, or future transaction-cost estimates. Those belong to later forecasting or execution layers. The sole purpose of $C_{i,t}$ is nuisance adjustment.

3 Inputs and notation

Symbol / field	Definition
i	Bond identifier.
t	Point-in-time scoring timestamp, usually end-of-day or a fixed intraday snapshot.
$g(i, t)$	Curve family / curve identifier used to value bond i at t (issuer, sector-rating, rating bucket, or another production curve family).
$Q_{g,t}(u)$	Existing par-implied survival curve for curve family g at time t . The target bond may have contributed to this curve.
$P_{i,t}^{mkt}$	Market price used for scoring, with a documented side convention.
$P_{i,t}^{curve}$	Bond price generated by the existing survival curve with zero bond-specific basis adjustment.
$\delta_{i,t}^{inc}$	Scalar bond-specific pricing basis to the <i>inclusive</i> survival curve, in spread bp. Positive means cheap.
$B_{i,t}^{own}$	Robust historical normal basis of bond i , estimated only from dates before t .
$q_{i,t}$	Bond-characteristics feature vector used in the nuisance regression.
$Y_{i,t}^C$	Training target for the characteristics regression; by definition $Y_{i,t}^C = B_{i,t}^{own}$.
$C_{i,t}$	Prediction of the characteristics model, in bp.
$B_{i,t}^{normal}$	Final estimate of the normal structural basis after blending own history and characteristics prediction.
$D_{i,t}$	Current abnormal RV dislocation after structural-basis removal.
$S_{i,t}$	Robust historical scale used to standardize $D_{i,t}$.
$Z_{i,t}$	Final standardized standalone RV score.

4 Step 1: compute the inclusive curve-pricing basis δ^{inc}

4.1 Preferred definition: exact repricing basis

Keep the existing survival curve fixed. Introduce a one-dimensional bond-specific credit-basis parameter b in the bond pricer. The exact interpretation of b must match the production analytics library (for example, an additive risky-discounting spread or another approved scalar spread add-on). Define $\delta_{i,t}^{inc}$ as the root of

$$\boxed{P_i(Q_{g(i,t),t}, b = \delta_{i,t}^{inc}; \text{bond terms}) = P_{i,t}^{mkt}}. \quad (3)$$

Use the sign convention

$$\boxed{\delta^{inc} > 0 \Rightarrow \text{bond is cheap to the curve,} \quad \delta^{inc} < 0 \Rightarrow \text{bond is rich to the curve.}} \quad (4)$$

The root solver must be bracketed and deterministic. Store both the root and the convergence status.

4.2 Linear approximation if an exact solver is unavailable

Let $CS01_{i,t} > 0$ denote the absolute price decline caused by a 1 bp spread widening. Then

$$P_{i,t}^{mkt} \approx P_{i,t}^{curve} - CS01_{i,t} \delta_{i,t}^{inc}, \quad (5)$$

so

$$\boxed{\delta_{i,t}^{inc} \approx \frac{P_{i,t}^{curve} - P_{i,t}^{mkt}}{CS01_{i,t}}}. \quad (6)$$

This approximation is acceptable for small dislocations. The exact root should be preferred when deviations are large or convexity/optionality is material.

4.3 Required implementation checks

1. Verify on at least 100 random bonds that lowering the market price while holding the curve fixed increases δ^{inc} .
2. Verify that a model price equal to the market price produces $\delta^{inc} \approx 0$.
3. Record curve ID, curve timestamp, bond analytics timestamp, price timestamp, and the pricing-side convention.
4. Never mix residuals computed from different pricing conventions in the same history without an explicit reset.

5 Step 2: estimate the bond's own historical normal basis B^{own}

5.1 Why this is necessary

A bond may persistently trade away from the curve because of issue size, age, coupon, seniority, callability, benchmark status, or liquidity. Such persistent basis is not necessarily a short-horizon alpha. The first defense is to estimate the bond's own normal location relative to the *same inclusive curve methodology*.

5.2 Default historical window

For every scoring date t , define a strictly lagged history

$$\mathcal{W}_{i,t} = \{u : t - L \leq u \leq t - k, \delta_{i,u}^{inc} \text{ is valid and comparable}\}. \quad (7)$$

A reasonable initial configuration is

$$L = 120 \text{ trading days,} \quad k = 5 \text{ trading days.} \quad (8)$$

The exclusion gap k prevents a newly developing dislocation from immediately contaminating the estimate of “normal.” These values are research parameters and must be validated, not hard-coded as universal constants.

5.3 Robust location estimate

Define

$$B_{i,t}^{own} = \text{median}_{u \in \mathcal{W}_{i,t}} \delta_{i,u}^{inc}. \quad (9)$$

Also retain

$$n_{i,t} = |\mathcal{W}_{i,t}| \quad (10)$$

and the robust historical scale

$$S_{i,t}^{raw} = 1.4826 \text{ median}_{u \in \mathcal{W}_{i,t}} |\delta_{i,u}^{inc} - B_{i,t}^{own}|. \quad (11)$$

The factor 1.4826 makes the MAD comparable to a standard deviation under a Gaussian reference distribution.

5.4 When B^{own} is invalid

Set $B_{i,t}^{own}$ to missing if any of the following is true:

- fewer than n_{min} valid observations are available;
- the bond changed curve family in a way that makes old residuals incomparable;
- a major contractual transformation occurred (exchange, restructuring, call exercise, seniority change, etc.);
- the pricing engine or basis definition changed materially and the old history was not recomputed consistently;
- the residual history is dominated by stale or failed pricing observations.

Do not silently fill missing B^{own} with zero.

6 Step 3: construct the characteristics-regression target

This is the central step.

6.1 Definition

For each training snapshot (i, t) where $B_{i,t}^{own}$ is valid, define

$$Y_{i,t}^C = B_{i,t}^{own}. \quad (12)$$

The target therefore has units of spread bp and means:

Based only on information available before date t , where did this bond normally sit relative to the existing inclusive survival curve?

6.2 Why today's raw residual is not the preferred target

A daily raw residual admits the decomposition

$$\delta_{i,t}^{inc} = \underbrace{B_{i,t}^{struct}}_{\text{persistent nuisance}} + \underbrace{D_{i,t}^{temp}}_{\text{temporary RV dislocation}} + \underbrace{\varepsilon_{i,t}}_{\text{measurement noise}}. \quad (13)$$

If the model is trained directly on $\delta_{i,t}^{inc}$, a sufficiently flexible model can start explaining D^{temp} , thereby subtracting the alpha itself. Training on the lagged robust location B^{own} makes the target much closer to the persistent nuisance component we actually want C to estimate.

6.3 Training-snapshot frequency

Do not create a training row every minute or every day when B^{own} changes only slowly. The overlapping windows would create many near-duplicate labels. A robust initial implementation is:

- one training snapshot per bond per week; or
- one snapshot every five trading days.

The scoring process may still run daily. Training and scoring frequencies need not be identical.

7 Step 4: define the characteristics vector $q_{i,t}$

7.1 Principle

Include variables that plausibly explain a *persistent structural basis*. Exclude variables whose main purpose is to forecast future returns or capture the current dislocation.

7.2 Recommended first-version features

Feature	Recommended transform / reason
Amount outstanding	$\log(1 + \text{amount})$; proxy for structural liquidity and benchmarkability.
Age since issue	$\log(1 + \text{age days})$; captures seasoning/new-issue effects.
Remaining maturity	Years or a small spline basis; controls maturity-related structural differences not already fully absorbed by curve construction.
Coupon	Coupon level, optionally interacted with price regime if required by analytics.
Seniority	Small categorical set: senior secured, senior unsecured, subordinated, etc.
Callable / putable	Contractual dummies; optionally time to next call if economically relevant.
Benchmark / on-the-run flag	If available and point-in-time.
Slow liquidity statistic	A lagged 20–60 day measure such as trade count, days-traded fraction, or median bid-ask; exclude the most recent few days.
Curve family / curve type	Use a small categorical variable or fit separate models by curve family.
Curve support count	Number of bonds supporting the curve; useful because inclusive-curve self-influence can differ with support.
Curve fit quality	Point-in-time RMSE / fit quality if available, strictly computed without future data.
Tenor support distance	Distance to nearest curve-support maturity or another simple extrapolation-risk metric.

7.3 Variables to exclude from q in version 1

Do not include today’s raw spread change, today’s curve residual, current order-flow imbalance, CDS move, equity return, ETF flow, dealer inventory, RFQ response skew, or any variable primarily intended to predict future return. Including those variables risks turning the nuisance model into an alpha model and subtracting genuine dislocation.

7.4 Point-in-time transformations and target-feature alignment

Every rolling feature must use data ending no later than $t - k_q$, where $k_q \geq 1$ day is preferred for features that could react contemporaneously to the same price dislocation. Example:

$$\text{Liquidity60}_{i,t} = \text{median liquidity statistic over } [t - 60, t - 5]. \quad (14)$$

This keeps “slow liquidity” structural rather than reactive.

There is an additional alignment rule. The target $B_{i,t}^{\text{own}}$ summarizes a historical window $[t - L, t - k]$. For a feature that changes materially across that same window (for example age, amount outstanding after tenders, or slow liquidity), the *training* feature should preferably be the median or mean feature state over the target window rather than the value observed only at the end of the window:

$$\tilde{q}_{i,t}^{\text{dyn}} = \text{median}_{u \in [t-L, t-k]} q_{i,u}^{\text{dyn}}. \quad (15)$$

Static contractual features can use their point-in-time value as of t . At production scoring time, use the current approved structural state (or its explicitly lagged slow version) because $C_{i,t}$ is intended to estimate the normal basis appropriate to today's bond state. This target-feature alignment avoids training a model on a 120-day average basis while presenting it with a materially different end-of-window characteristic value.

8 Step 5: build the training table

Each row in the characteristics-model training set must contain exactly one point-in-time observation:

$$(i, t, Y_{i,t}^C, q_{i,t}, n_{i,t}, S_{i,t}^{raw}, curve_id, timestamps). \quad (16)$$

8.1 Eligibility rules

A row is eligible only if:

1. $Y_{i,t}^C$ is based exclusively on dates earlier than t ;
2. $n_{i,t} \geq n_{min}$;
3. the bond terms are valid at t ;
4. all curve and price inputs were available at t ;
5. there is no known data-methodology break inside the historical window unless the history was recomputed consistently;
6. the target is finite and inside a broad data-quality bound chosen before model fitting.

8.2 Target-precision weight

Because a median estimated from 100 observations is more reliable than one estimated from 15 observations, assign each row a training weight. A useful approximation to the standard error of a sample median under a Gaussian reference model is

$$SE(B_{i,t}^{own}) \approx 1.2533 \frac{S_{i,t}^{raw}}{\sqrt{n_{i,t}}}. \quad (17)$$

Define a capped precision weight

$$v_{i,t} = \text{clip} \left(\frac{1}{SE(B_{i,t}^{own})^2 + s_0^2}, v_{min}, v_{max} \right), \quad (18)$$

where s_0 prevents extremely quiet bonds from receiving infinite weight. Simpler count-based weights are acceptable for an initial prototype, but precision weights are only slightly more work and are statistically cleaner.

9 Step 6: fit the characteristics model

9.1 Recommended first production model

Use a robust regularized linear model. Let $\phi(q)$ denote the processed feature vector after transformations, standardization, and one-hot encoding of small categorical variables. Fit

$$\hat{\gamma}_t = \arg \min_{\gamma_0, \gamma} \sum_{(i,u) \in \mathcal{T}_t} v_{i,u} \rho_{\kappa} (Y_{i,u}^C - \gamma_0 - \phi(q_{i,u})^\top \gamma) + \lambda \|\gamma\|_2^2, \quad (19)$$

where ρ_{κ} is Huber loss and $\mathcal{T}_t$ contains training dates strictly before the model fit date.

The fitted structural-basis prediction is

$$C_{i,t} = \hat{\gamma}_{0,t} + \phi(q_{i,t})^\top \hat{\gamma}_t. \quad (20)$$

9.2 Why start with this model

The target is a nuisance adjustment, not the final alpha. The first version should favor stability, interpretability, and low variance. A linear/Huber/ridge model makes it easy to see whether amount outstanding, age, seniority, curve support, or slow liquidity are producing sensible basis adjustments. XGBoost may be tested later, but only after the linear model is stable and the incremental out-of-sample reduction in structural-basis error is clear.

9.3 Separate models versus pooled model

Use one of two architectures:

- **Separate models:** fit separate C models for materially different curve families (for example issuer-curve residuals versus sector-rating-curve residuals).
- **Pooled model:** include curve type and support metrics in q and allow a small number of interactions.

Do not pool incompatible residual definitions merely to increase sample size.

10 Step 7: fit point-in-time and validate without leakage

10.1 Rolling fit schedule

A simple production schedule is monthly model refitting with daily scoring. For a model used during month M , fit it using only training snapshots dated before the first day of M .

10.2 Validation split

Use chronological validation. For example:

- training: first 24 months;
- validation: next 6 months;
- test: subsequent untouched period.

Within research, roll these windows forward. Because the same bond appears many times, also run an issuer-held-out or bond-held-out robustness test.

10.3 Metrics

The nuisance model should be evaluated on structural-basis prediction, not future returns. Report:

$$MAE_C = \text{median} |Y^C - C|, \quad (21)$$

$$RMSE_C = \sqrt{\mathbb{E}[(Y^C - C)^2]}, \quad (22)$$

$$Bias_C = \mathbb{E}[Y^C - C]. \quad (23)$$

Also report these by curve type, rating, sector, maturity bucket, amount-outstanding bucket, age bucket, and history length.

10.4 Sanity tests

A valid C model should usually satisfy:

- prediction distributions are much smoother than daily raw residuals;
- new bonds receive sensible predictions even before enough own history exists;
- predictions do not react sharply to a one-day spread dislocation;
- feature coefficients / partial effects are stable across nearby training windows;
- adding alpha-like fast variables should not be necessary for good structural-basis prediction.

11 Step 8: combine own history and C into the normal structural basis

11.1 Why blend

For a seasoned bond, its own history should dominate. For a new bond, the characteristics prediction should dominate. A blend provides this behavior continuously.

11.2 Preferred precision-weighted blend

Let

$$V_{i,t}^{own} = SE(B_{i,t}^{own})^2. \quad (24)$$

Estimate a characteristics-model prediction variance $V_{i,t}^C$ from out-of-sample residuals, at least by broad bucket (curve type, rating/liquidity bucket). Then define

$$w_{i,t}^{own} = \frac{V_{i,t}^C}{V_{i,t}^{own} + V_{i,t}^C}, \quad (25)$$

with $w_{i,t}^{own} = 0$ when $B_{i,t}^{own}$ is unavailable. The normal basis is

$$B_{i,t}^{normal} = w_{i,t}^{own} B_{i,t}^{own} + (1 - w_{i,t}^{own}) C_{i,t}. \quad (26)$$

This is the inverse-variance combination of two approximately unbiased estimates.

11.3 Simpler fallback

If prediction variances are not yet implemented, use

$$w_{i,t}^{own} = \frac{n_{i,t}}{n_{i,t} + \lambda_n}, \quad (27)$$

with $w_{i,t}^{own} = 0$ when $B_{i,t}^{own}$ is missing. This is easier but less statistically precise. Treat λ_n as a validation parameter.

12 Step 9: compute the abnormal dislocation and final score

The current abnormal relative-value dislocation is

$$D_{i,t} = \delta_{i,t}^{inc} - B_{i,t}^{normal}. \quad (28)$$

Positive D means unusually cheap after structural-basis adjustment; negative D means unusually rich.

12.1 Robust scale

For seasoned bonds, estimate the scale from past abnormal deviations using only historical model versions / point-in-time normal-basis estimates:

$$S_{i,t} = 1.4826 \text{ median}_{u \in \mathcal{W}_{i,t}^S} |D_{i,u} - \text{median}(D_{i,\cdot})|. \quad (29)$$

For an initial implementation where historical point-in-time C predictions have not yet been backfilled, it is acceptable to use $S_{i,t}^{raw}$ as the denominator, but mark that as version 0 and replace it once historical adjusted residuals are available.

12.2 Final score

$$Z_{i,t} = \frac{D_{i,t}}{S_{i,t} + \epsilon}. \quad (30)$$

Interpretation:

$$Z > 0 \Rightarrow \text{cheap}, \quad Z < 0 \Rightarrow \text{rich}, \quad |Z| \Rightarrow \text{extremeness relative to history}. \quad (31)$$

13 Worked numerical example

Assume today's inclusive survival-curve pricing basis is

$$\delta_{i,t}^{inc} = +18 \text{ bp.} \quad (32)$$

From the lagged 120-day window excluding the last 5 days, the bond has

$$B_{i,t}^{own} = +7 \text{ bp}, \quad n_{i,t} = 92. \quad (33)$$

The characteristics model, based on amount outstanding, age, seniority, callability, maturity, slow liquidity, and curve support, predicts

$$C_{i,t} = +5 \text{ bp.} \quad (34)$$

Suppose the own-history estimate is relatively precise and the precision weight gives

$$w^{own} = 0.80. \quad (35)$$

Then

$$B^{normal} = 0.8(7) + 0.2(5) = 6.6 \text{ bp.} \quad (36)$$

The abnormal dislocation is

$$D = 18 - 6.6 = 11.4 \text{ bp.} \quad (37)$$

If the robust historical scale is

$$S = 4.75 \text{ bp,} \quad (38)$$

then

$$\boxed{Z = 11.4/4.75 = +2.40.} \quad (39)$$

The meaning is not “the bond is exactly 11.4 bp cheap to an uncontaminated fair-value curve.” The meaning is: *relative to the same inclusive survival-curve methodology and after removing the best estimate of the bond's normal structural basis, today's deviation is approximately 2.4 robust historical scale units cheap.*

14 End-to-end algorithm

14.1 Algorithm A: build historical inclusive residuals

1. Freeze a curve snapshot and bond-security-master snapshot at time t .
2. Obtain $P_{i,t}^{mkt}$ under the approved price-side convention.
3. Reprice the bond on the existing survival curve.
4. Solve for $\delta_{i,t}^{inc}$ (exact root preferred; CS01 approximation allowed as a fallback).
5. Persist $\delta_{i,t}^{inc}$ together with curve ID, curve type, curve support count, curve fit quality, price timestamp, analytics version, and root-solver status.
6. Repeat for every eligible bond-date. Never rewrite old rows without a versioned recomputation.

14.2 Algorithm B: build B^{own} and the regression target

For each eligible training snapshot (i, t) :

1. Select historical residuals in $[t - L, t - k]$.
2. Remove observations rendered incomparable by methodology or curve-family changes.
3. Require at least n_{min} valid observations.
4. Compute $B_{i,t}^{own} = \text{median}(\delta^{inc})$.

5. Compute $S_{i,t}^{raw} = 1.4826 \text{ MAD}(\delta^{inc})$.
6. Set the characteristics-model target $Y_{i,t}^C = B_{i,t}^{own}$.
7. Build $q_{i,t}$ using point-in-time structural characteristics and slow lagged liquidity variables.
8. Compute target precision weight $v_{i,t}$.
9. Store one immutable training row.

14.3 Algorithm C: fit the characteristics model

1. Choose a model-fit date T_{fit} .
2. Select only training rows with snapshot date $< T_{fit}$.
3. Fit preprocessing transforms on the training data only.
4. Fit robust ridge / Huber regression with sample weights v .
5. Tune λ and Huber threshold using chronological validation only.
6. Freeze model artifact, preprocessing artifact, feature schema, category mapping, and model version.
7. Measure out-of-sample residual variance by broad bucket to obtain V^C for blending.

14.4 Algorithm D: score a bond today

1. Obtain today's $\delta_{i,t}^{inc}$ from the existing curve and market price.
2. Compute today's $q_{i,t}$ with the same feature definitions used in training.
3. Predict $C_{i,t}$ from the frozen characteristics model.
4. Compute $B_{i,t}^{own}$ from history if enough comparable observations exist.
5. Compute $w_{i,t}^{own}$ from precision estimates (or the count-based fallback).
6. Compute $B_{i,t}^{normal}$.
7. Compute $D_{i,t} = \delta_{i,t}^{inc} - B_{i,t}^{normal}$.
8. Compute $S_{i,t}$ from historical adjusted residuals.
9. Compute $Z_{i,t} = D_{i,t} / (S_{i,t} + \epsilon)$.
10. Persist every intermediate quantity. A production score is invalid if any required version/timestamp lineage is missing.

15 Reference pseudocode

```
# INPUTS AT DATE t
# delta_inc[i,t] : current inclusive survival-curve pricing basis, bp
# history       : past delta_inc rows for this bond
# q[i,t]        : point-in-time structural characteristics
# C_model       : model fitted only on earlier dates

# 1. Own historical normal basis
hist = history[(date >= t-L) & (date <= t-k) & comparable]
if len(hist) >= n_min:
    B_own = median(hist.delta_inc)
    S_raw = 1.4826 * median(abs(hist.delta_inc - B_own))
    se_own = 1.2533 * S_raw / sqrt(len(hist))
else:
```

```

B_own = NaN
S_raw = NaN
se_own = inf

# 2. Characteristics prediction
C = C_model.predict(q[i,t]) # bp

# 3. Blend normal basis
if isfinite(B_own):
    V_own = se_own**2
    V_C = lookup_oos_prediction_variance(curve_type, rating_bucket, liquidity_bucket)
    w_own = V_C / (V_own + V_C)
else:
    w_own = 0.0

B_normal = w_own * B_own + (1.0 - w_own) * C

# 4. Abnormal dislocation
D = delta_inc[i,t] - B_normal

# 5. Final score
S = historical_adjusted_MAD(i, t) # strictly lagged, same score definition
Z = D / max(S, scale_floor)

```

16 Training-table schema

A minimum table for model fitting should contain:

Column	Description
snapshot_date	Date on which the training row is defined.
bond_id	Stable bond identifier.
issuer_id	Used for grouped validation; not necessarily a model feature.
curve_id / curve_type	Exact curve family used for δ^{inc} .
y_C_bp	$B_{i,t}^{own}$; this is the regression target.
n_hist	Number of observations used in B^{own} .
S_raw_bp	Historical MAD scale.
weight_target	Precision weight $v_{i,t}$.
log_amount_out	Structural feature.
log_age_days	Structural feature.
remaining_maturity	Structural feature.
coupon	Structural feature.
seniority	Small categorical feature.
callable / putable	Contractual feature.
slow_liquidity	Lagged slow liquidity feature.
curve_support_count	Number of bonds supporting the curve.
curve_fit_rmse	Point-in-time curve fit quality.
tenor_support_distance	Curve support / extrapolation diagnostic.
feature_version	Immutable feature-definition version.
target_version	Immutable B^{own} construction version.

17 Minimum test suite before research results are trusted

17.1 Unit tests

1. **Pricing-basis sign:** lower market price $\Rightarrow$ larger positive δ^{inc} .
2. **No-lookahead history:** maximum date used in $B_{i,t}^{own}$ is $\leq t - k$.
3. **Target identity:** verify exactly $Y_{i,t}^C = B_{i,t}^{own}$ for every training row.

4. **Missing-history rule:** insufficient history never becomes zero basis.
5. **Feature causality:** every rolling feature's maximum source timestamp satisfies its lag rule.
6. **Model causality:** scoring date is later than the model's maximum training date.
7. **Blend limits:** $V^{own} \rightarrow 0 \Rightarrow w^{own} \rightarrow 1$; $V^{own} \rightarrow \infty \Rightarrow w^{own} \rightarrow 0$.
8. **Score sign:** $\delta^{inc} > B^{normal}$ gives positive D and positive Z .

17.2 Integration tests

1. Pick 20 bonds and reconstruct every intermediate quantity by hand / notebook from raw tables.
2. Re-run the pipeline after shuffling rows; outputs must be identical.
3. Re-run with all data after scoring date deleted; outputs must be identical.
4. Hold out an issuer and verify the model can still generate sensible C predictions from structural characteristics.
5. For new issues, verify $w^{own} \approx 0$ and that the score still computes from C plus an approved cross-sectional scale fallback if used.
6. After sufficient history accumulates, verify the blend transitions smoothly from C -dominated to own-history-dominated.

18 Failure modes and explicit remediation

18.1 Failure mode: the characteristics model eats the alpha

Symptom: $C_{i,t}$ moves sharply on the same day as the bond dislocation. Remedy: remove fast/current variables, increase feature lags, and verify the training target is B^{own} rather than daily δ^{inc} .

18.2 Failure mode: target leakage through the historical median

Symptom: the historical window includes dates after the snapshot or includes the recent dislocation. Remedy: enforce $u \leq t - k$ in code and unit-test source timestamps.

18.3 Failure mode: method changes contaminate history

Symptom: residual distribution jumps when the curve engine or pricing convention changes. Remedy: recompute history under one version or reset the historical window at the methodology break.

18.4 Failure mode: curve self-influence changes over time

Symptom: the number or maturity support of curve constituents changes sharply, altering inclusive-residual shrinkage. Remedy: include curve support count / support distance / fit quality in q , split model by curve type if necessary, and reject scores during extreme support breaks.

18.5 Failure mode: new issues have no scale

Symptom: C exists but $S_{i,t}$ is unavailable. Remedy: use a separately validated cross-sectional scale model or bucket-level robust scale for new issues. Do not fabricate a zero denominator. Once enough own history exists, transition to bond-specific scale.

19 Recommended implementation order for an agent

The safest implementation sequence is deliberately incremental.

1. **Data discovery only.** Locate existing survival-curve snapshots, market prices, bond terms, curve identifiers, curve-support metadata, and historical residual-capable analytics. Produce a data dictionary; do not code the model yet.
2. **Pricing-basis prototype.** Implement δ^{inc} for 20 bonds on one date. Verify sign and units manually.
3. **Historical residual table.** Generate δ^{inc} for a small 3-month sample. Validate lineage and missingness.
4. **Own-basis prototype.** Implement B^{own} , S^{raw} , and n with an explicit $[t - L, t - k]$ window. Unit-test no look-ahead.
5. **Training target table.** Create weekly snapshot rows with $Y^C = B^{own}$. Stop and inspect target distributions before adding features.
6. **Feature table.** Add only structural / slow features. Verify each feature's source timestamp.
7. **Linear baseline.** Fit weighted ridge first. Inspect signs, residuals, and subgroup errors.
8. **Robust baseline.** Upgrade to Huber + ridge if outliers materially influence the fit.
9. **Point-in-time validation.** Run chronological and issuer-held-out validation. Do not proceed if the model only works in random CV.
10. **Blend implementation.** Add V^{own} , V^C , w^{own} and B^{normal} . Test limiting cases.
11. **Final score.** Add D and Z . Verify long/short sign convention on hand-selected bonds.
12. **Backfill and monitoring.** Backfill historical point-in-time C , B^{normal} , D , and Z before any return-prediction study.

20 Final formula stack

The entire component can be summarized by six equations.

1. Inclusive curve basis

$$P_i(Q_{g,t}, \delta_{i,t}^{inc}) = P_{i,t}^{mkt}. \quad (40)$$

2. Own normal basis

$$B_{i,t}^{own} = \text{median}_{u \in [t-L, t-k]} \delta_{i,u}^{inc}. \quad (41)$$

3. Characteristics-model target

$$\boxed{Y_{i,t}^C = B_{i,t}^{own}}. \quad (42)$$

4. Characteristics prediction

$$\boxed{C_{i,t} = \hat{m}_t(q_{i,t})}. \quad (43)$$

5. Normal structural basis

$$\boxed{B_{i,t}^{normal} = w_{i,t}^{own} B_{i,t}^{own} + (1 - w_{i,t}^{own}) C_{i,t}}. \quad (44)$$

6. Abnormal RV score

$$Z_{i,t} = \frac{\delta_{i,t}^{inc} - B_{i,t}^{normal}}{S_{i,t} + \epsilon}. \quad (45)$$

The critical interpretation is:

$C_{i,t}$ estimates the persistent basis expected from bond characteristics. The final score is the current inclusive curve deviation minus the estimated normal structural basis, scaled by historical residual variability. The model is intentionally designed so that C removes nuisance structure without being trained on future returns or current short-horizon alpha variables.